Mu (Henry) Ha

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Eligible for TN visa sponsorship under USMCA (Engineer / Computer Systems Analyst categories)

Manufacturing: DFM/DFA, GD&T, Tolerance Analysis, SAP BOM & routings, cost analysis, Quality Control, SPC (Cp/Cpk), Yield Analysis, Lean Manufacturing, Test Plan Development, Root Cause Analysis
Automation & Data: Python, SQL, Bash/Shell, C/C++, REST APIs, TCP/IP, AWS, PostgreSQL, Pandas, Plotly
Hardware Integration: Simulink, Arduino, PLCs, Sensor Integration, Dynamic Modeling, PID, Sensor Fusion
Engineering Tools: MATLAB, SolidWorks, Siemens NX, AutoCAD, ANSYS, Git/GitHub, CI/CD, Docker

EXPERIENCE

Brilliant Automation (Capstone) — Apr 2025 – Jun 2025
Systems Engineer — Vancouver, BC

- Built a sensor data validation pipeline processing multi-rate telemetry and waveform inputs to generate interpretable health scores across 12 machine metrics for production monitoring.
- Implemented leakage-free expanding-window CV (5 splits, N=3,045) with structured OOF evaluation, achieving R^2=0.39 vs. negative baseline to ensure test results reflect true deployment performance.
- Deployed monitoring dashboards to track performance metrics, anomaly trends, and qualification auditability.

Weir Minerals (Internship) — Sep 2019 – Apr 2020
Manufacturing Engineer — Surrey, BC

- Estimated production costs and created BOMs & routings in SAP to support production planning and control.
- Conducted mechanical validation testing (gasket compression, deformation) and led root cause investigations to resolve quality deviations and ensure spec compliance.
- Improved cost estimation workflow by ~30% through data standardization, enhancing traceability & planning.
- Ensured compliance with manufacturing quality standards and supported validation documentation.

Weir Minerals (Internship) — Jan 2019 – Apr 2019
Design Engineer — Toronto, ON

- Designed mechanical assemblies with GD&T to ensure manufacturability and tolerance robustness.
- Collaborated with fabrication teams to resolve tolerance stack-ups and support production validation.
- Performed design verification including material selection, seal geometry checks, and stress validation.
- Generated controlled engineering drawings and documentation to support manufacturing releases.

PROJECTS

Manufacturing Test Automation Platform — Dec 2025 – Jan 2026

- Developed a TCP/IP-based DUT simulator to emulate manufacturing test communication, control, and faults.
- Designed structured acceptance and qualification test plans with defined coverage mapping, retry logic, and step-level validation criteria tied to product requirements.
- Implemented an automated test framework supporting multi-SN execution, structured JSON logging, fault injection (timeouts, drift, intermittent failure), and traceable result storage.
- Built yield analytics modules (FPY, FTY, Pareto, stratification) and deployed a Dockerized, CI-enabled validation pipeline with auto-generated qualification reports.

Robotic Manipulator System — Aug 2023 – Sep 2023

- Integrated motors, sensors, and Arduino control for closed loop hardware validation testing.
- Developed manual and automated control modes to validate trajectory accuracy and repeatability.
- Performed system identification and parameter tuning to resolve instability and coupling effects.
- Executed structured troubleshooting and root cause analysis to improve control robustness.
- Validated performance under varying load and motion profiles to ensure stable operation.

EDUCATION

University of British Columbia — Sep 2024 – Jul 2025
Master of Data Science
University of Waterloo — Sep 2018 – Apr 2024
B.Sc. Mechanical Engineering *(Honours, Co-op) AI Option, Management Science Option*